EXP NO: 04 STUDY OF LOCAL AREA NETWORK USING A SWITCH & ETHERNAL CABLES

AIM:

To set up and configure a Local Area Network (LAN) using a switch and Ethernel cables, and verify communication between connected computer.

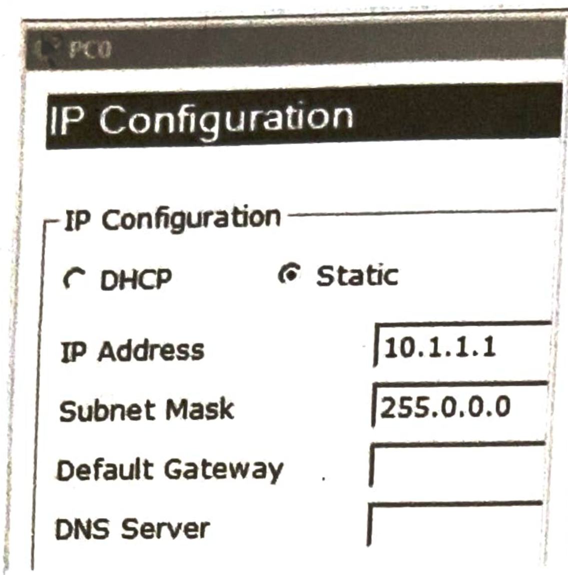
INTRODUCTION: A Local Area Network (LAN) is a network that connects computers and other devices within a limited geographical area such a laberatory, school, and home. It enables users to share resources like files, printers and internet connections.

LANs provide reliable and internet connections, secure and cost-effective communication between devices.

A network switch acts as the central connecting devices, allowing all connected computers to communicate with each other at high speed.

PROCEDURE:

- Plan the network by arranging 4 computers, one network switch and Ethernal cables.
- Connect each computer to the switch using Ethernel cables.
- open control panel → Network → Internet → Network connection.
- Right-click Ethernet → properties → TCP / IPV4 → properties.



PC0

IP Configuration

IP Configuration

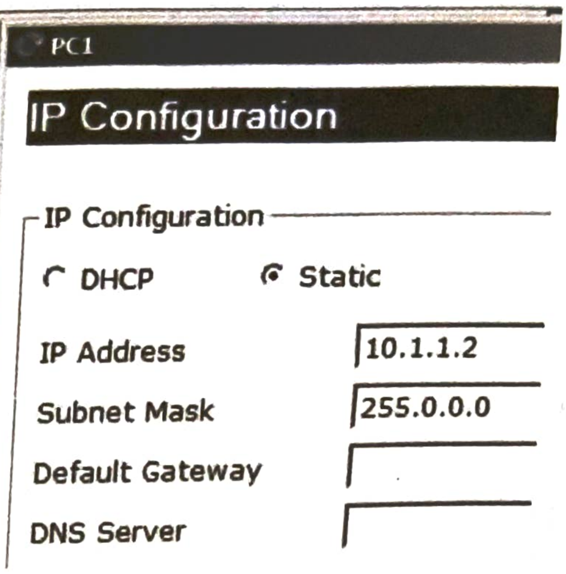
○ DHCP ● Static

IP Address: 10.1.1.1

Subnet Mask: 255.0.0.0

Default Gateway:

DNS Server:



PC1

IP Configuration

IP Configuration

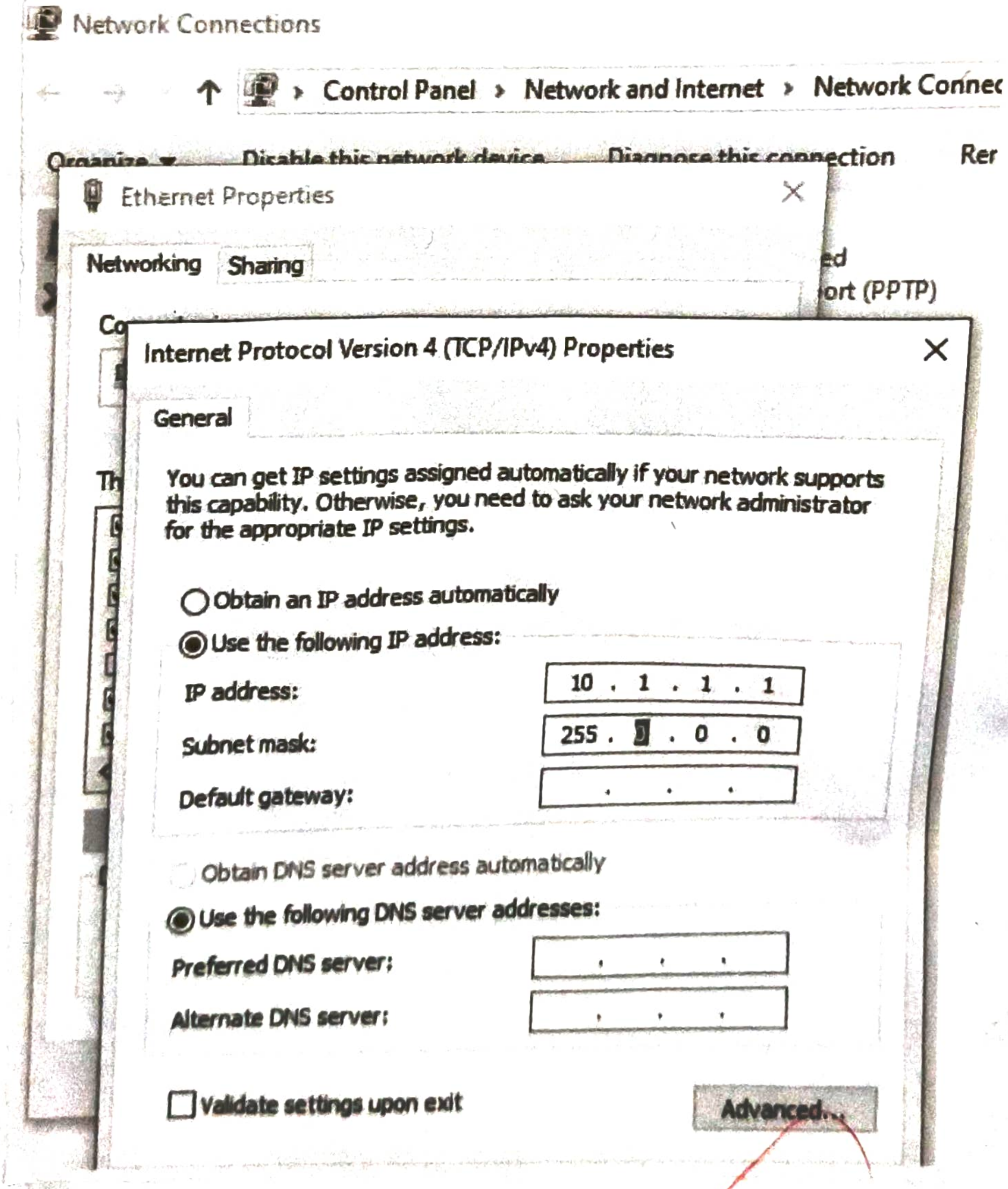
○ DHCP ● Static

IP Address: 10.1.1.2

Subnet Mask: 255.0.0.0

Default Gateway:

DNS Server:



Network Connections

← → ↑ › Control Panel › Network and Internet › Network Connec

Organize ▾
Disable this network device
Diagnose this connection
Rer
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Ethernet Properties

Networking
Sharing
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Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

○ Obtain an IP address automatically

● Use the following IP address:

IP address: 10 . 1 . 1 . 1

Subnet mask: 255 . 0 . 0 . 0

Default gateway: . . .

○ Obtain DNS server address automatically

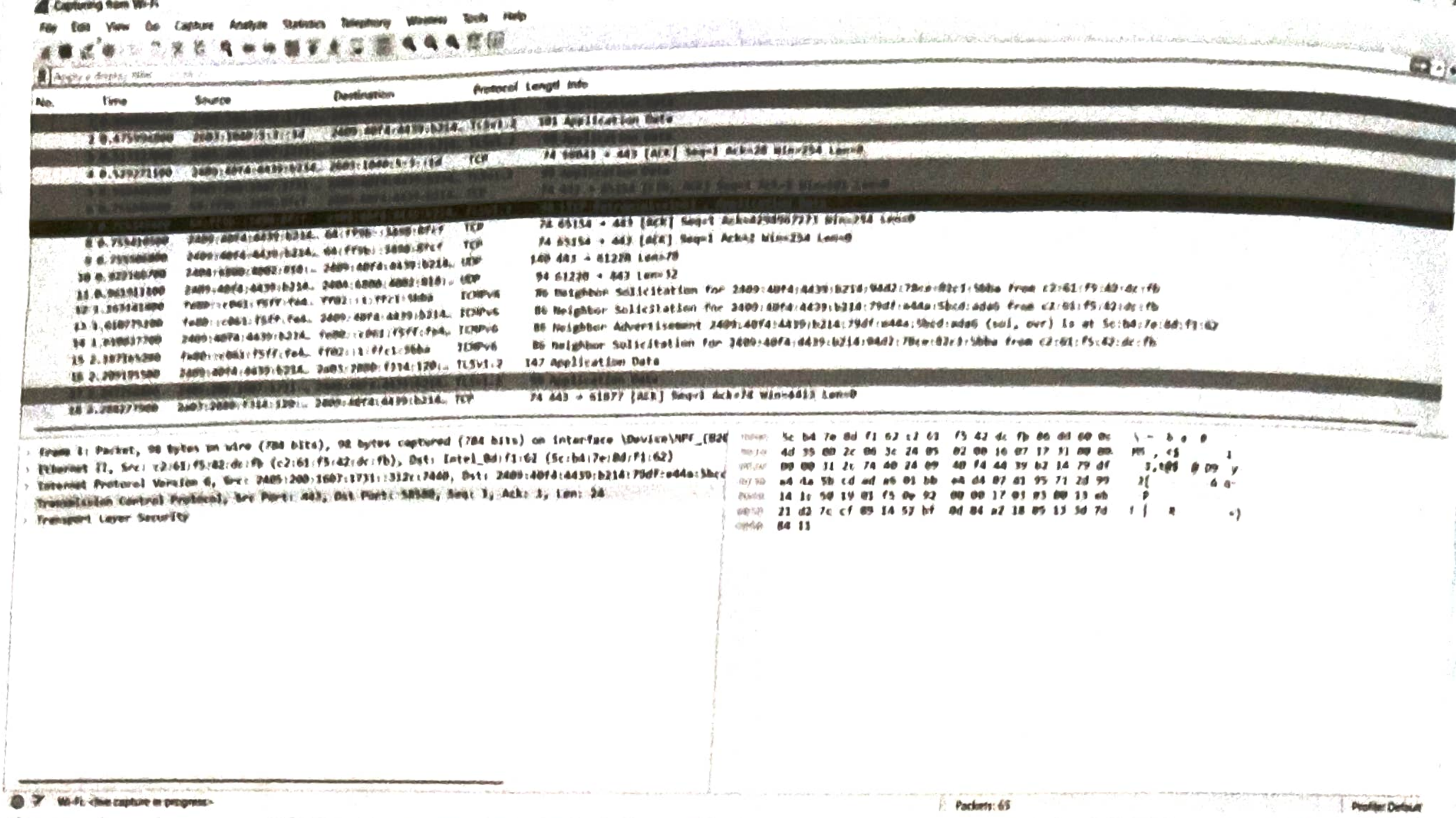
● Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

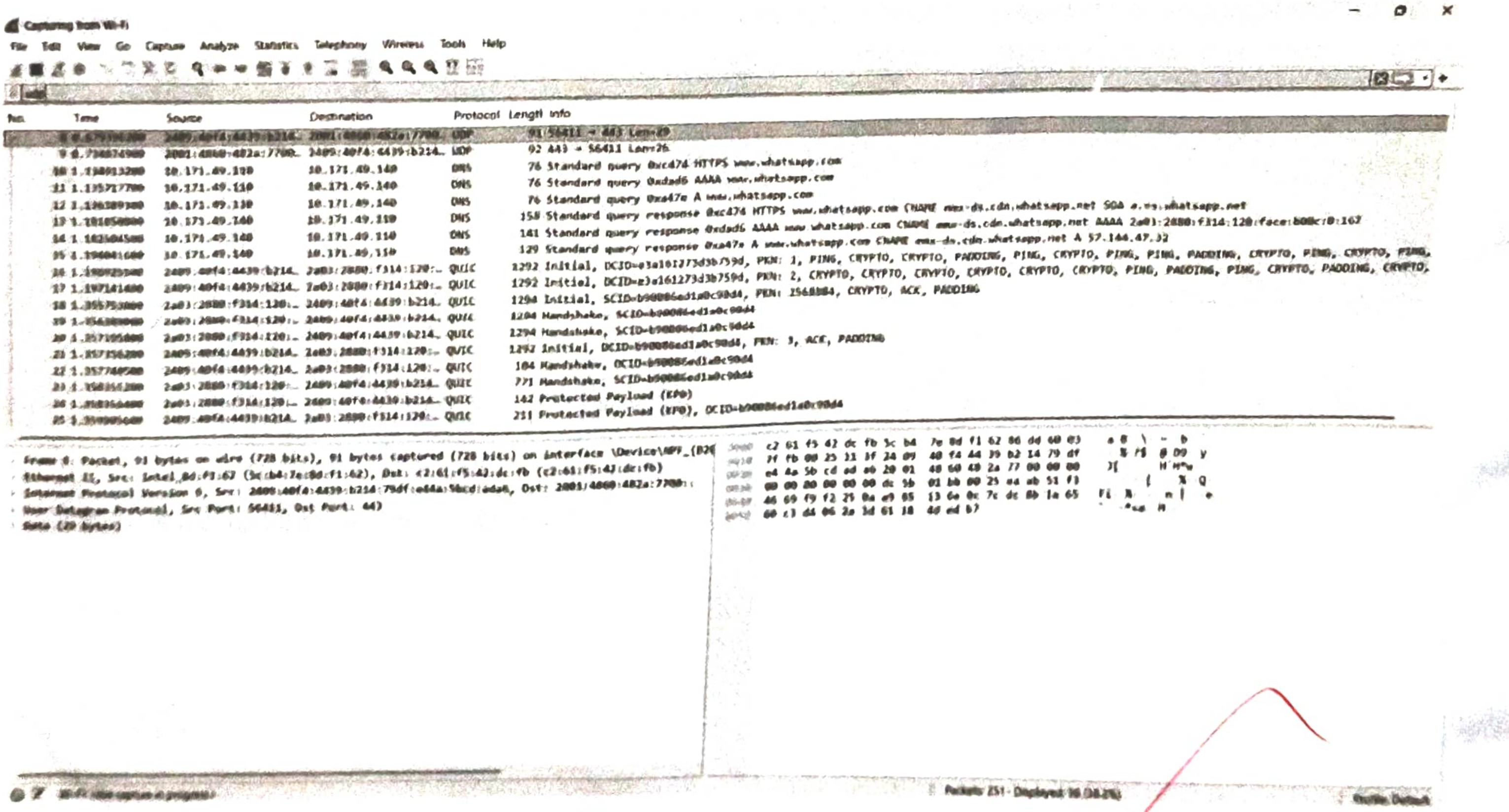
✓


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter
Wi-Fi: <live capture in progress>
Packets: 65
Profile: Default

```
No.  Time           Source                   Destination              Protocol  Length Info
 1   [illegible]                                                      TLSv1.2   [illegible]
 2   0.475[illegible]   2401:1600:5:1::14    2409:40f4:4439:b214..    TLSv1.2   101 Application Data
 3   [illegible]                                                      TLSv1.2   98 Application Data
 4   0.529271100    2409:40f4:4439:b214..    2401:1600:5:1::14        TCP       74 58041 → 443 [ACK] Seq=1 Ack=28 Win=254 Len=0
 5   [illegible]                                                      TLSv1.2   [illegible]
 6   [illegible]                                                      TCP       [illegible]
 7   [illegible]                                                      TCP       74 65154 → 443 [ACK] Seq=1 Ack=4294967271 Win=254 Len=0
 8   0.755410500    2409:40f4:4439:b214..    64:ff9b::3490:8fcf       TCP       74 65154 → 443 [ACK] Seq=1 Ack=42 Win=254 Len=0
 9   0.755[illegible]   2409:40f4:4439:b214..    64:ff9b::3490:8fcf   TCP       140 443 → 61220 Len=78
10   0.927160700    2404:6800:4002:810:..    2409:40f4:4439:b214..    UDP       94 61220 → 443 Len=32
11   0.963917100    2409:40f4:4439:b214..    2404:6800:4002:810:..    UDP       86 Neighbor Solicitation for 2409:40f4:4439:b214:94d2:78ce:02c1:5bba from c2:61:f5:42:dc:fb
12   1.367141400    fe80::c061:f5ff:fe4..    ff02::1:ffc1:5bba        ICMPv6    86 Neighbor Solicitation for 2409:40f4:4439:b214:79df:e44a:5bcd:ada6 from c2:61:f5:42:dc:fb
13   1.610779100    fe80::c061:f5ff:fe4..    2409:40f4:4439:b214..    ICMPv6    86 Neighbor Advertisement 2409:40f4:4439:b214:79df:e44a:5bcd:ada6 (sol, ovr) is at 5c:b4:7e:8d:f1:62
14   1.910837700    2409:40f4:4439:b214..    fe80::c061:f5ff:fe4..    ICMPv6    86 Neighbor Solicitation for 2409:40f4:4439:b214:94d2:78ce:02c1:5bba from c2:61:f5:42:dc:fb
15   2.187165200    fe80::c061:f5ff:fe4..    ff02::1:ffc1:5bba        ICMPv6    147 Application Data
16   2.209195500    2409:40f4:4439:b214..    2a03:2880:f314:120:..    TLSv1.2   98 Application Data
17   [illegible]    2405:200:1607:1731:..    2409:40f4:4439:b214..    TLSv1.2
18   2.288277900    2a03:2880:f314:120:..    2409:40f4:4439:b214..    TCP       74 443 → 61877 [ACK] Seq=1 Ack=74 Win=4413 Len=0
```

```
> Frame 1: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF_{B2
> Ethernet II, Src: c2:61:f5:42:dc:fb (c2:61:f5:42:dc:fb), Dst: Intel_8d:f1:62 (5c:b4:7e:8d:f1:62)
> Internet Protocol Version 6, Src: 2405:200:1607:1731::312c:7460, Dst: 2409:40f4:4430:b214:79df:e44a:5bcd
> Transmission Control Protocol, Src Port: 443, Dst Port: 58580, Seq: 1, Ack: 1, Len: 24
> Transport Layer Security
```

```
0000  5c b4 7e 8d f1 62 c2 61  f5 42 dc fb 86 dd 60 0c   \~ b a `
0010  4d 35 00 2c 06 3c 24 05  02 00 16 07 17 31 00 00   M5 , <$      1
0020  00 00 31 2c 74 60 24 09  40 f4 44 39 b2 14 79 df     1,t`$ @ D9  y
0030  e4 4a 5b cd ad a6 01 bb  e4 d4 07 41 95 71 2d 99    J[        A q-
0040  14 1c 50 19 01 f5 0e 92  00 00 17 03 03 00 13 eb     P
0050  21 d2 7c cf 89 14 57 bf  8d 84 a2 18 05 13 5d 7d   ! |   W        ]}
0060  84 11
```


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
Wi-Fi: <live capture in progress>
Packets: 251 · Displayed: 96 (38.2%)
Profile: Default

```
No.  Time           Source                   Destination              Protocol  Length Info
 8   0.679196200    2409:40f4:4439:b214..    2001:4860:482a:7700..    UDP       91 56411 → 443 Len=29
 9   0.734874900    2001:4860:482a:7700..    2409:40f4:4439:b214..    UDP       92 443 → 56411 Len=26
10   1.130013200    10.171.49.110            10.171.49.140            DNS       76 Standard query 0xc474 HTTPS www.whatsapp.com
11   1.135717700    10.171.49.110            10.171.49.140            DNS       76 Standard query 0xdad6 AAAA www.whatsapp.com
12   1.136389300    10.171.49.110            10.171.49.140            DNS       76 Standard query 0xa47e A www.whatsapp.com
13   1.181050000    10.171.49.140            10.171.49.110            DNS       158 Standard query response 0xc474 HTTPS www.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net SOA a.ns.whatsapp.net
14   1.182504500    10.171.49.140            10.171.49.110            DNS       141 Standard query response 0xdad6 AAAA www.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net AAAA 2a03:2880:f314:120:face:b00c:0:167
15   1.196041600    10.171.49.140            10.171.49.110            DNS       129 Standard query response 0xa47e A www.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net A 57.144.47.32
16   1.196925900    2409:40f4:4439:b214..    2a03:2880:f314:120:..    QUIC      1292 Initial, DCID=e3a161273d3b759d, PKN: 1, PING, CRYPTO, CRYPTO, PADDING, PING, CRYPTO, PING, PING, PADDING, CRYPTO, PING, CRYPTO, PING,
17   1.197141400    2409:40f4:4439:b214..    2a03:2880:f314:120:..    QUIC      1292 Initial, DCID=e3a161273d3b759d, PKN: 2, CRYPTO, CRYPTO, CRYPTO, CRYPTO, CRYPTO, CRYPTO, PING, PADDING, PING, CRYPTO, PADDING, CRYPTO,
18   1.355753000    2a03:2880:f314:120:..    2409:40f4:4439:b214..    QUIC      1294 Initial, SCID=b90086ed1a0c90d4, PKN: 1568884, CRYPTO, ACK, PADDING
19   1.356380000    2a03:2880:f314:120:..    2409:40f4:4439:b214..    QUIC      1294 Handshake, SCID=b90086ed1a0c90d4
20   1.357105000    2a03:2880:f314:120:..    2409:40f4:4439:b214..    QUIC      1294 Handshake, SCID=b90086ed1a0c90d4
21   1.357156200    2409:40f4:4439:b214..    2a03:2880:f314:120:..    QUIC      1292 Initial, DCID=b90086ed1a0c90d4, PKN: 3, ACK, PADDING
22   1.357740500    2409:40f4:4439:b214..    2a03:2880:f314:120:..    QUIC      104 Handshake, DCID=b90086ed1a0c90d4
23   1.358356200    2a03:2880:f314:120:..    2409:40f4:4439:b214..    QUIC      771 Handshake, SCID=b90086ed1a0c90d4
24   1.358356400    2a03:2880:f314:120:..    2409:40f4:4439:b214..    QUIC      142 Protected Payload (KP0)
25   1.359905600    2409:40f4:4439:b214..    2a03:2880:f314:120:..    QUIC      211 Protected Payload (KP0), DCID=b90086ed1a0c90d4
```

```
> Frame 8: Packet, 91 bytes on wire (728 bits), 91 bytes captured (728 bits) on interface \Device\NPF_{B2
> Ethernet II, Src: Intel_8d:f1:62 (5c:b4:7e:8d:f1:62), Dst: c2:61:f5:42:dc:fb (c2:61:f5:42:dc:fb)
> Internet Protocol Version 6, Src: 2409:40f4:4439:b214:79df:e44a:5bcd:ada6, Dst: 2001:4860:482a:7700::
> User Datagram Protocol, Src Port: 56411, Dst Port: 443
> Data (29 bytes)
```

```
0000  c2 61 f5 42 dc fb 5c b4  7e 8d f1 62 86 dd 60 03    a B \ ~ b    `
0010  7f fb 00 25 11 3f 24 09  40 f4 44 39 b2 14 79 df     % ?$ @ D9  y
0020  e4 4a 5b cd ad a6 20 01  48 60 48 2a 77 00 00 00    J[     H`H*w
0030  00 00 00 00 00 00 dc 5b  01 bb 00 25 a4 ab 51 f1          [   %  Q
0040  46 69 f9 12 25 0a e9 05  13 6e 0c 7c dc 8b 1a 65   Fi  %    n |   e
0050  60 c3 d4 06 2a 3d 61 18  4d ed b7                  `   *=a M
```

* Select use the following IP address and assign
* -PC1 : 10:1.1.1, Subnet Mask : 225.0.0.0
* -PC2 : 10.1.1.2, Subnet Mask : 255.0.0.0
* -PC3 : 10.1.1.3, Subnet Mask : 255.0.0.0
* -PC4 : 10.1.1.4, Subnet Mask : 255.0.0.0

* Configuration the switch by connecting it to a computer accessing its web interface and assigining the switch IP as 10.1.1.5 with 255.0.0.0.

RESULT:

Thus, A Local Area Network (LAN) was successfully set up and configured using a network switch and Ethernet cables.